

RONCHI DEI LEGIONARI, Italy

WMO: 161080

Lat: 45.828N

Lon: 13.472E

Elev: 11

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-3.2	-13.8	1.1	-1.6	-11.3	1.4	0.3	10.6	4.1	9.2	4.3	1.4	50	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	33.2	22.9	31.9	22.3	30.2	21.5	24.4	30.3	23.5	29.4	22.6	28.4	2.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.6	27.5	21.8	16.5	26.6	20.8	15.5	25.7	73.8	30.3	70.0	29.1	66.8	28.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.9	7.3	5.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-7.9	35.5	2.8	1.7	-9.9	36.7	-11.5	37.7	-13.1	38.6	-15.1	39.8		
			-9.4	26.1	2.6	1.0	-11.2	26.8	-12.8	27.4	-14.2	27.9	-16.1	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	4.5	5.3	9.0	12.9	17.5	21.6	23.6	23.3	18.8	14.3	9.7	5.3		
	DBStd	7.48	3.01	3.09	3.00	2.98	2.79	2.96	2.58	2.87	3.00	3.27	3.71	3.22		
	HDD10.0	569	173	132	53	8	0	0	0	0	0	6	49	148		
	HDD18.3	2127	429	364	291	163	50	6	1	2	30	128	259	403		
	CDD10.0	1984	2	2	21	96	232	348	423	413	264	139	41	3		
	CDD18.3	499	0	0	0	1	23	104	166	157	44	4	0	0		
	CDH23.3	4724	0	0	1	20	211	984	1646	1525	321	17	0	0		
	CDH26.7	1703	0	0	0	2	38	333	645	617	67	1	0	0		
(14) Wind	WSAvg	2.0	1.7	2.0	2.2	2.1	2.0	1.9	2.1	2.0	2.0	1.9	1.8	1.7		
(15) Precipitation	PrecAvg	1187	75	70	78	102	95	95	91	90	121	148	136	93		
	PrecMax	1639	266	241	258	235	243	237	159	215	302	323	435	231		
	PrecMin	848	7	3	0	9	12	22	34	13	37	15	24	6		
	PrecStd	207	59	66	65	57	52	42	39	53	69	86	92	62		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.2	21.0	25.9	29.9	34.0	35.0	35.8	30.8	25.2	19.8	14.9		
		MCWB	11.0	9.5	12.2	16.5	20.4	23.3	24.7	23.4	21.2	17.2	14.4	12.2		
	2%	DB	12.1	14.0	18.6	23.0	27.7	31.9	33.0	33.6	28.5	23.0	17.8	13.0		
		MCWB	9.2	8.6	11.5	15.1	19.2	22.2	23.1	22.6	20.2	17.5	14.4	10.1		
	5%	DB	10.8	12.1	16.6	21.1	25.8	30.1	31.7	31.9	26.9	21.2	16.2	11.7		
		MCWB	8.1	8.0	10.6	13.8	17.9	21.3	22.4	22.2	19.6	16.8	13.4	8.8		
	10%	DB	9.2	10.9	14.9	19.2	23.9	28.8	30.1	30.1	25.0	20.0	15.1	10.2		
		MCWB	7.0	7.4	9.8	13.1	17.1	20.8	21.7	21.5	18.7	16.1	12.4	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.2	13.6	17.4	21.8	24.9	25.9	25.5	23.3	20.5	16.6	13.5		
		MCDB	13.3	13.9	18.0	23.6	27.0	30.3	32.2	30.6	27.7	21.9	18.2	14.5		
	2%	WB	10.3	10.1	12.4	15.9	20.1	23.3	24.6	24.3	21.8	19.1	15.2	11.3		
		MCDB	11.3	12.1	16.5	21.2	26.1	29.4	30.7	30.1	25.9	21.4	16.9	12.5		
	5%	WB	8.8	9.0	11.6	14.7	18.8	22.3	23.5	23.4	20.7	18.1	14.1	9.6		
		MCDB	9.8	11.2	15.1	19.4	24.0	28.5	29.5	29.3	24.8	20.3	16.1	10.5		
	10%	WB	7.5	7.9	10.8	13.7	17.8	21.4	22.6	22.5	19.7	16.9	13.1	8.4		
		MCDB	8.7	10.0	13.7	18.0	22.4	27.1	28.4	28.4	23.7	19.1	14.7	9.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.3	10.2	10.1	10.3	10.9	11.4	11.5	10.6	9.7	8.1	8.2		
		MCDBR	9.5	12.2	13.8	13.8	13.4	13.1	13.2	13.8	12.6	11.3	8.8	10.0		
		MCWBR	6.7	7.9	8.3	7.4	6.6	6.0	6.0	5.9	6.3	6.7	6.0	7.3		
		MCWBR	5.9	8.9	10.7	11.7	11.5	12.0	12.3	12.4	10.4	8.1	6.6	6.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.342	0.368	0.416	0.460	0.466	0.480	0.468	0.459	0.440	0.419	0.375	0.338		
		taud	2.400	2.317	2.192	2.098	2.130	2.142	2.185	2.225	2.240	2.263	2.334	2.431		
		Ebn at Noon	749	795	802	799	808	797	801	792	770	723	693	712		
		Edh at Noon	74	96	125	150	150	149	141	131	119	101	79	66		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.36	2.23	3.47	4.56	5.56	6.29	6.39	5.49	4.00	2.51	1.44	1.19		
		RadStd	0.20	0.37	0.45	0.53	0.57	0.38	0.36	0.46	0.36	0.28	0.22	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.45	N/A	N/A	+0.72	+0.44	N/A	-80	-126	+88	+54

Nomenclature: See separate page